1. **Course number and name**
   CHE 454 Molecular and Macromolecular Engineering

2. **Credits, contact hours and categorization**
   Credit hours: 3. Contact hours: 3 lecture hours per week.

3. **Instructor's or course coordinator's name**
   Vivek Sharma

4. **Text book, title, author, publisher, year**

(1) Polymer Chemistry, Second Edition, Paul C. Hiemenz, Timothy P. Lodge, CRC Press, 2007.
(2) Polymer science, A Materials Science Handbook, Volume I & II, Jenkins, A.D. North, Holland Publishing Co., Amsterdam and London; American Elsevier Publishing Company, New York, 1972.

5. **Specific course information**
   a. Brief description of the content of the course (catalog description)
   Advanced course in polymer science and engineering. Polymerization, polydispersity, molecular configuration, solution properties, thermodynamics, glass and rubbery states, crystallization, viscoelasticity, elastic properties, multiphase systems.
   Prerequisites or co-requisites
   CHE 330.
   b. Indicate whether a required, elective, or selected elective.
   Elective.

6. **Specific goals for the course**
   a. Specific outcomes of instruction.

- Understand the fundamental and advanced concepts used in macromolecular systems for chemical engineering applications
- Learn the detailed methods used to synthesize and characterize macromolecular systems
- Understand the structural, viscoelastic, mechanical and thermal properties of macromolecules and their characterization methods.
- Reinforce the ability to communicate effectively through individual papers and presentations in class.

a. Explicitly indicate which of the student outcomes are addressed by the course.

		1	2	3	4
(7)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				x
(7)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(7)	an ability to communicate effectively with a range of audiences.				x
(7)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(7)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(7)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			x	
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				x

7. Brief list of topics to be covered

- Introduction to Plastics and Polymer materials
- Molecular Structures, Microstructures, and Polymerization
- Plastics Manufacturing
- Introduction to Composites
- Composites: Materials and Manufacturing
- Thermoplastics
- Composite Processing
- Introduction to 3D printing
- Printing in plastic, metal, glass, wood, concrete and other materials
- Bioprinting
- Environmental Aspects of Polymer-based Materials and Material processing